Site Suitability Analysis For The Construction Of PHC In Araku Valley Using AHP

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INTRODUCTION:

- Hospitals have become the most critical facilities where people go for health-related issues and to improve their health standards. Many countries and institutions give great focus to hospital site selection. Choosing the most suitable location from several alternative sites is demanding and complex.

- **Primary Health Centre** (**PHC**s) are state-owned rural and urban health care facilities in India. They are part of the government-funded public health system in India. As of our study area which has less number of PHC's there is need to select a site suitable for constructing a PHC.

- Site selection is a decision-making process that requires the weighing of criteria and the evaluation and classification of alternatives . It is necessary to establish more hospitals to enhance healthcare facilities Choosing optimal hospital locations hinges on several factors. This project validates the effectiveness of the analytical hierarchy process (AHP), the most widely applied model in the Multiple-Criteria Decision Analysis (MCDA).

Suitability analysis, or site selection, is a GIS-based process used to determine the appropriateness of a given area for a particular use. The basic premise of suitability analysis is that each aspect of the landscape has intrinsic characteristics that are in some degree either suitable or unsuitable for the activities being planned. Different data layers are created which correlate to various physical, cultural, or economic factors that would be important for site selection analysis. The results are often displayed on a map that is used to highlight areas from high to low suitability.

Workflow:

1. Define the problem and prepare the criteria data.
2. Transform the input values you've generated to a common suitability scale so the criteria can be compared.
3. Rank criteria from most suitable to least suitable and combine the rankings to create a suitability map.
4. Use the suitability map to locate sites.

Description of Site Evaluation Criteria **for constructing a PHC:**

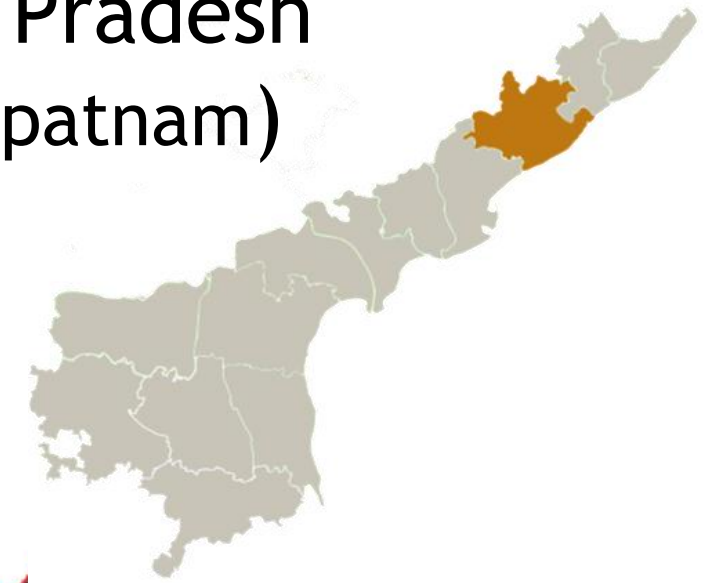
1) **Conditions of land** – flat, no waterbody and must be a barren land
2) **Surface slope** - elevation should be less than 3%
3) **Road distance** – 1-2km from the roads
4) **Rock type** - Charnockite> khondalite> migmatite
5) **Buffer** - 500 meters around the existing hospitals.

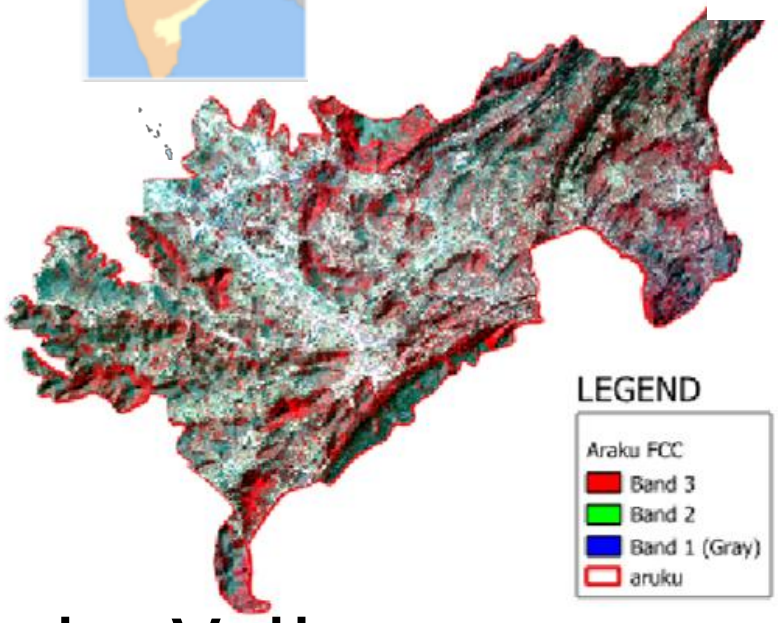
Study Area: Araku

INDIA



Andhra Pradesh
(Visakhapatnam)





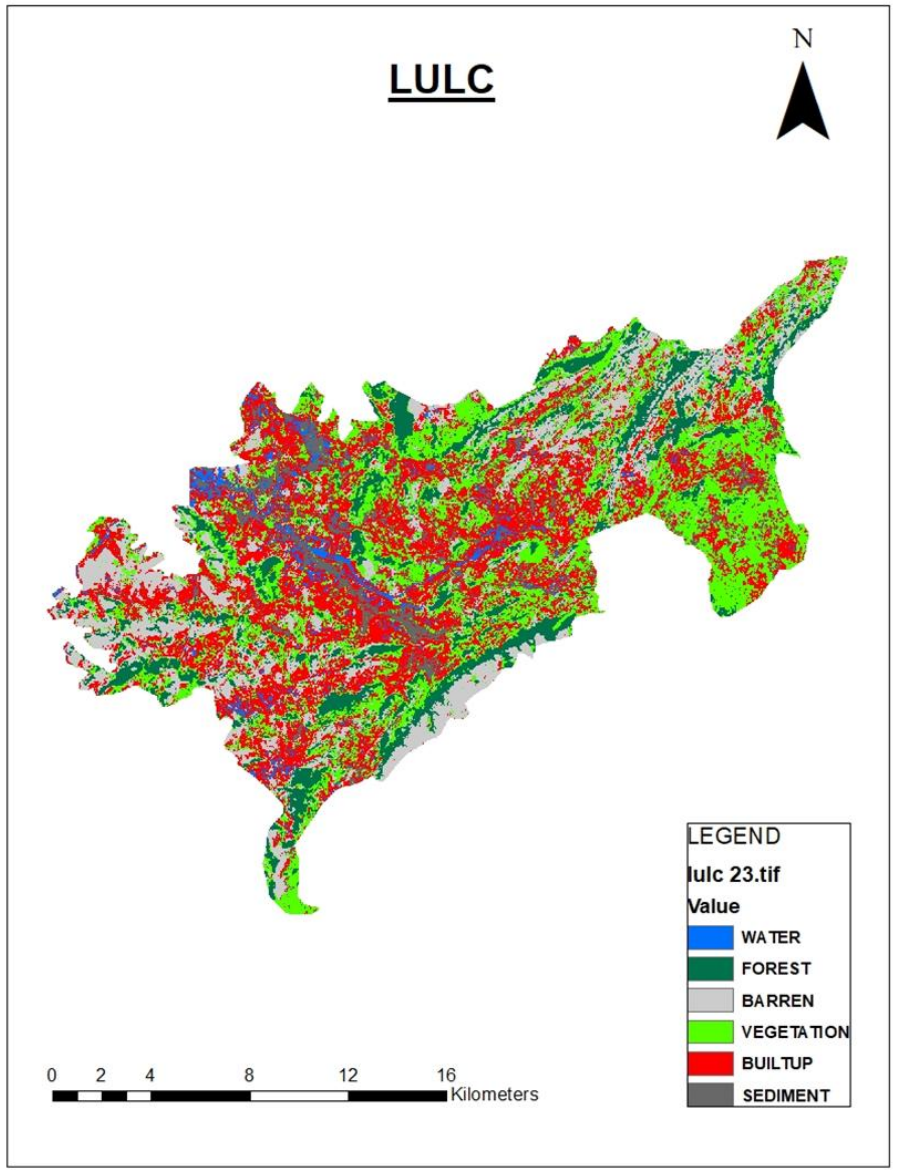
Araku Valley FCC Landsat 8 TM

Araku

The study area is Araku Valley in Visakhapatnam District of Andhra Pradesh State situated between 18.3333°N latitude and 82.8667°E longitude. Total area of Araku valley mandal is 264 km^2 including 262.44 km^2 rural area and 1.62 km^2 urban area. Araku Valley mandal has a population of 56,674 people. Out of total population, 88.15% of population lives in Urban area and 11.85% lives in Rural area. Araku Valley mandal has a population density of 215 inhabitants per square kilometer.

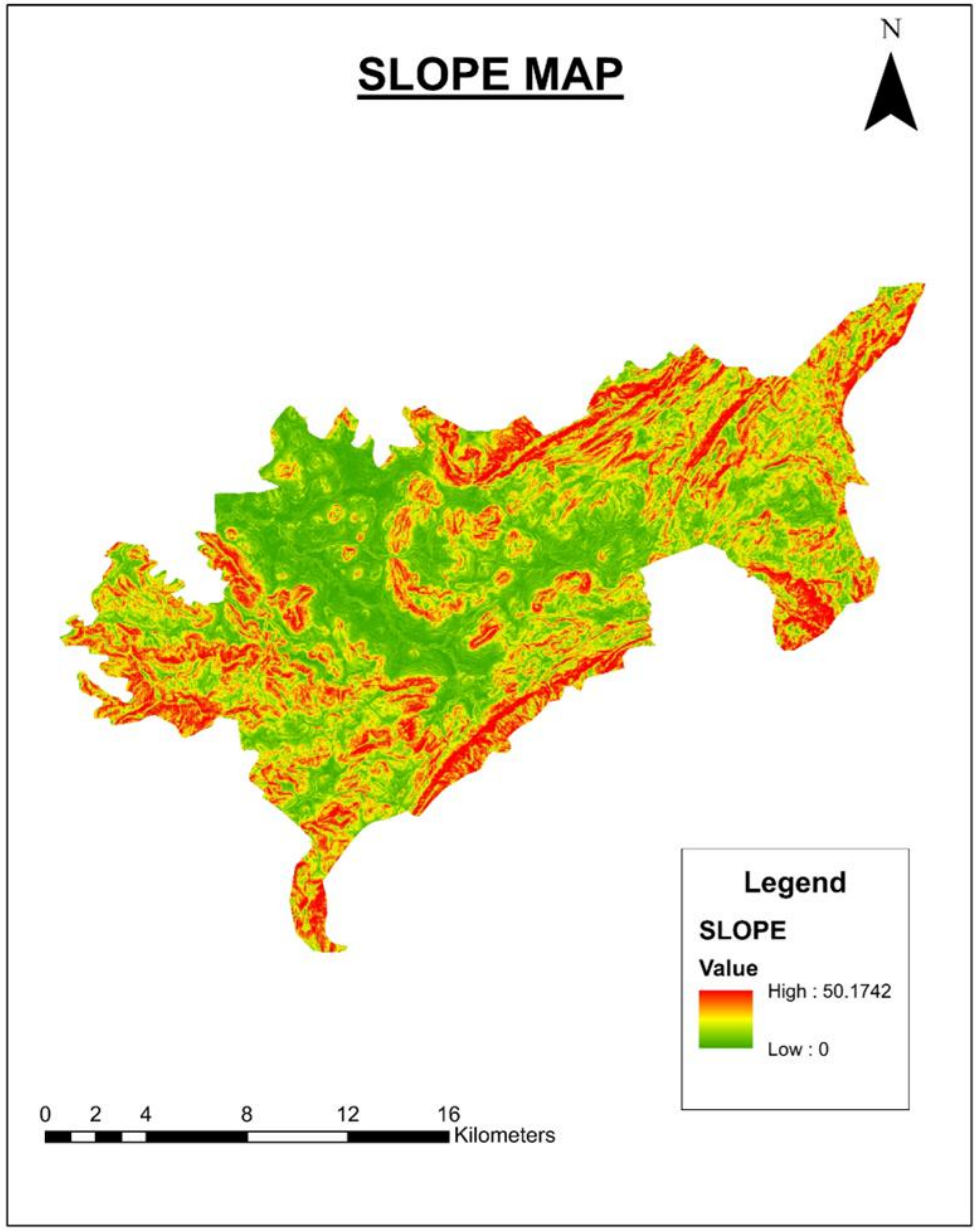
Total Villages	215
Population	56674
Male Population	48.51%
Female Population	51.49%
Children Population	7224
Area	264.06 km2
Population density/km2	215

Data Collection:



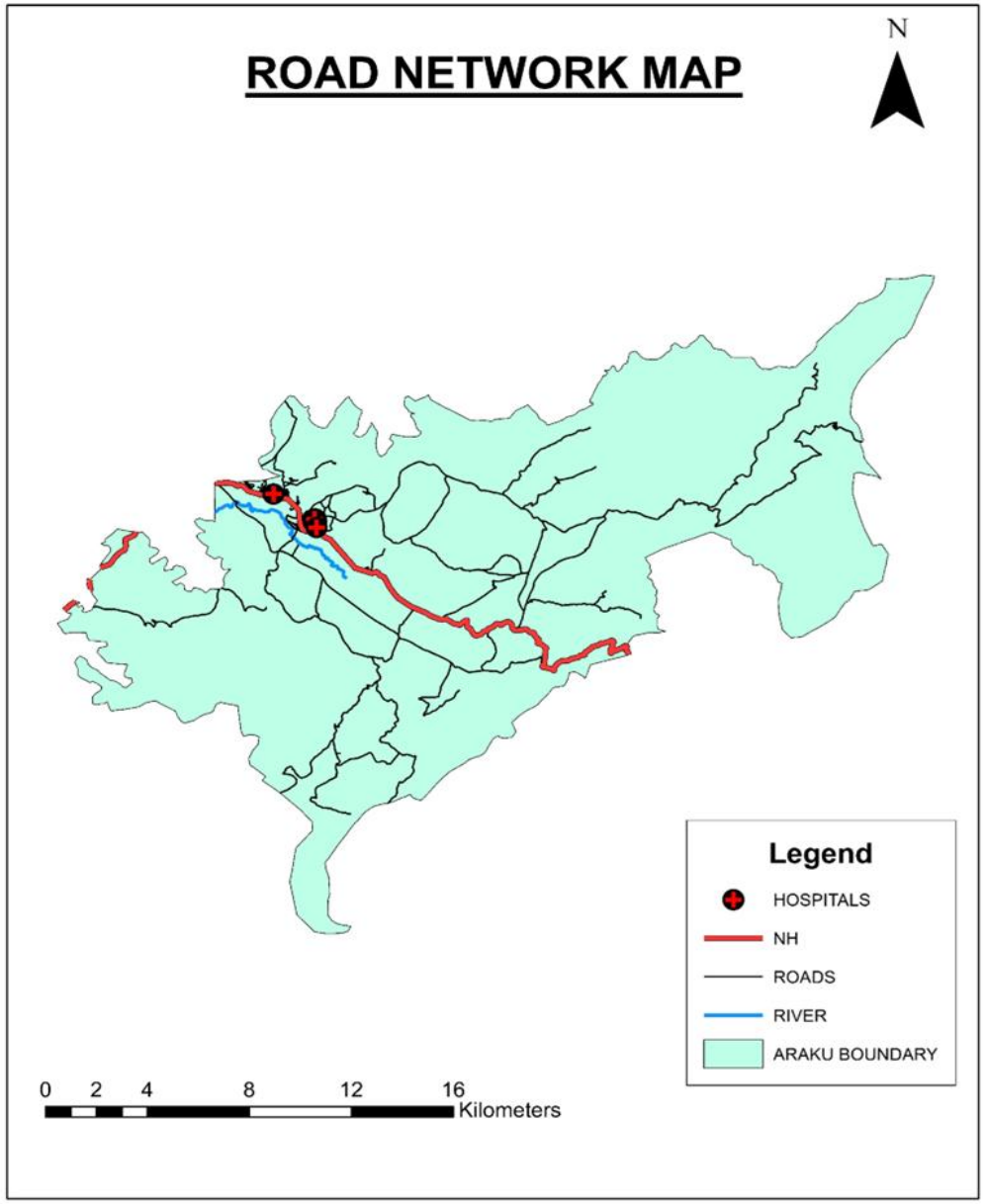
Land use represents an essential parameter in the choice of land. Barren land, built-up areas, and areas with low vegetation density characterized by high susceptibility to flooding. The land use map, as shown in Figure, was classified into six classes according to the degree of susceptibility. Land Use Land Cover Map created by doing supervised classified using Landsat 8 TM satellite image.

Data Source: USGS Earth Explorer



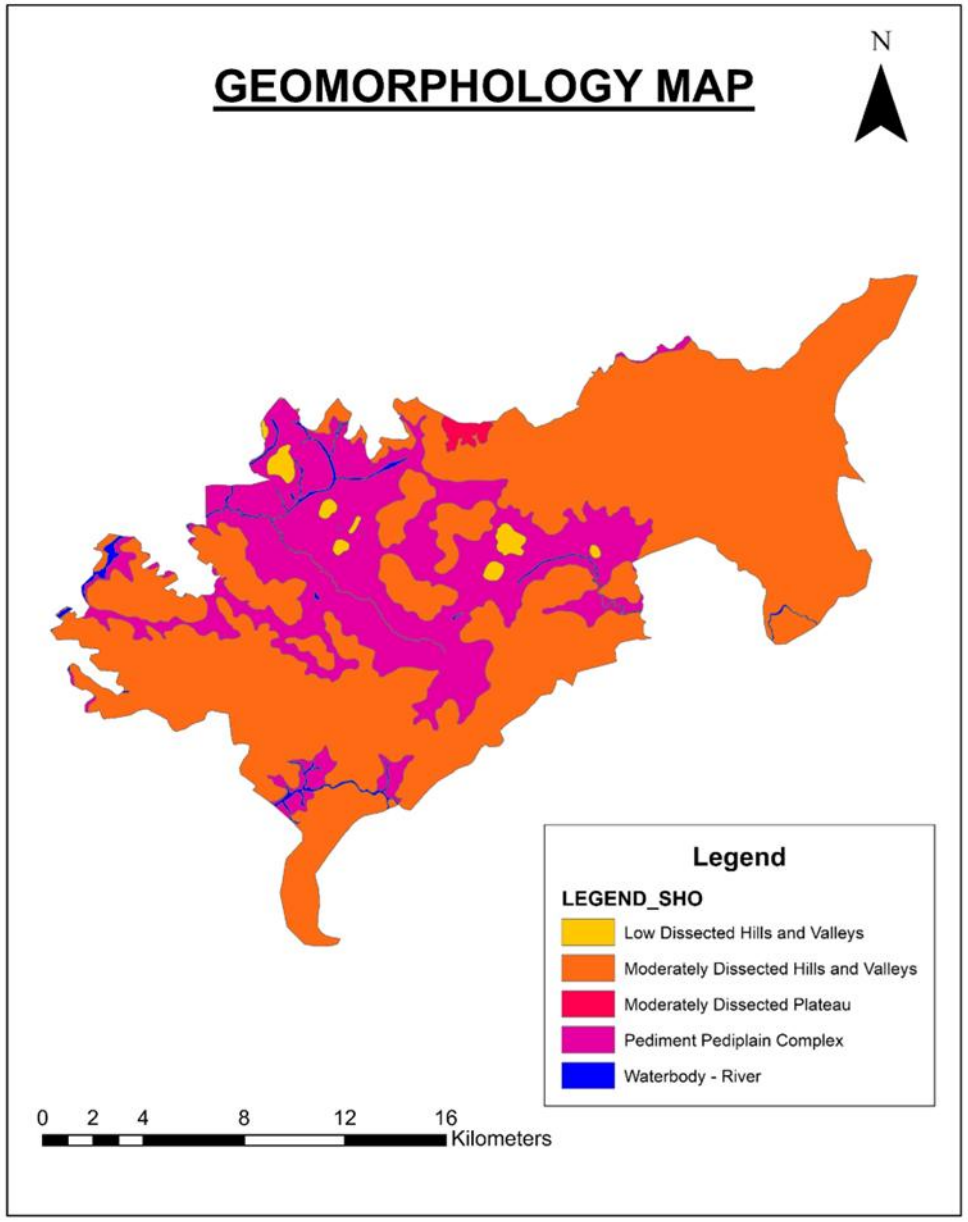
Slope plays a significant role in the construction of hospital site. The intrinsic factor allows the control of runoff and determine surface water stagnation during occasional rains .The slope is expressed in degrees, generated from a digital elevation model of required study area.

Data Source: Cartosat DEM 1arc sec(BHUVAN portal)



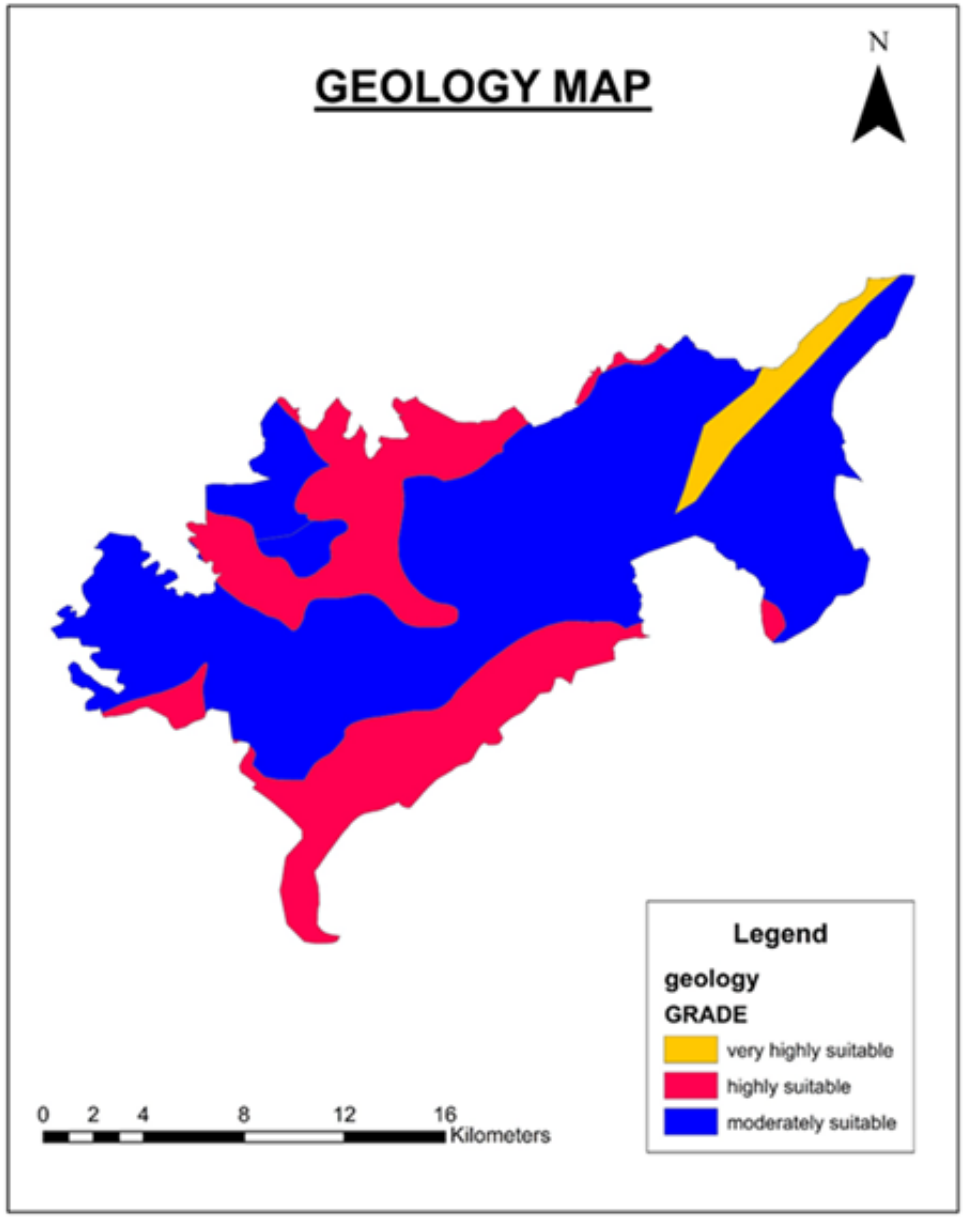
The proximity of roads is another essential factor determines that the level of difficulty to transport on site. Proximity to roads reduces transportation costs and time.
Road Network Map is obtained by digitizing the Araku satellite image.

Data Source : HCMGIS plugin



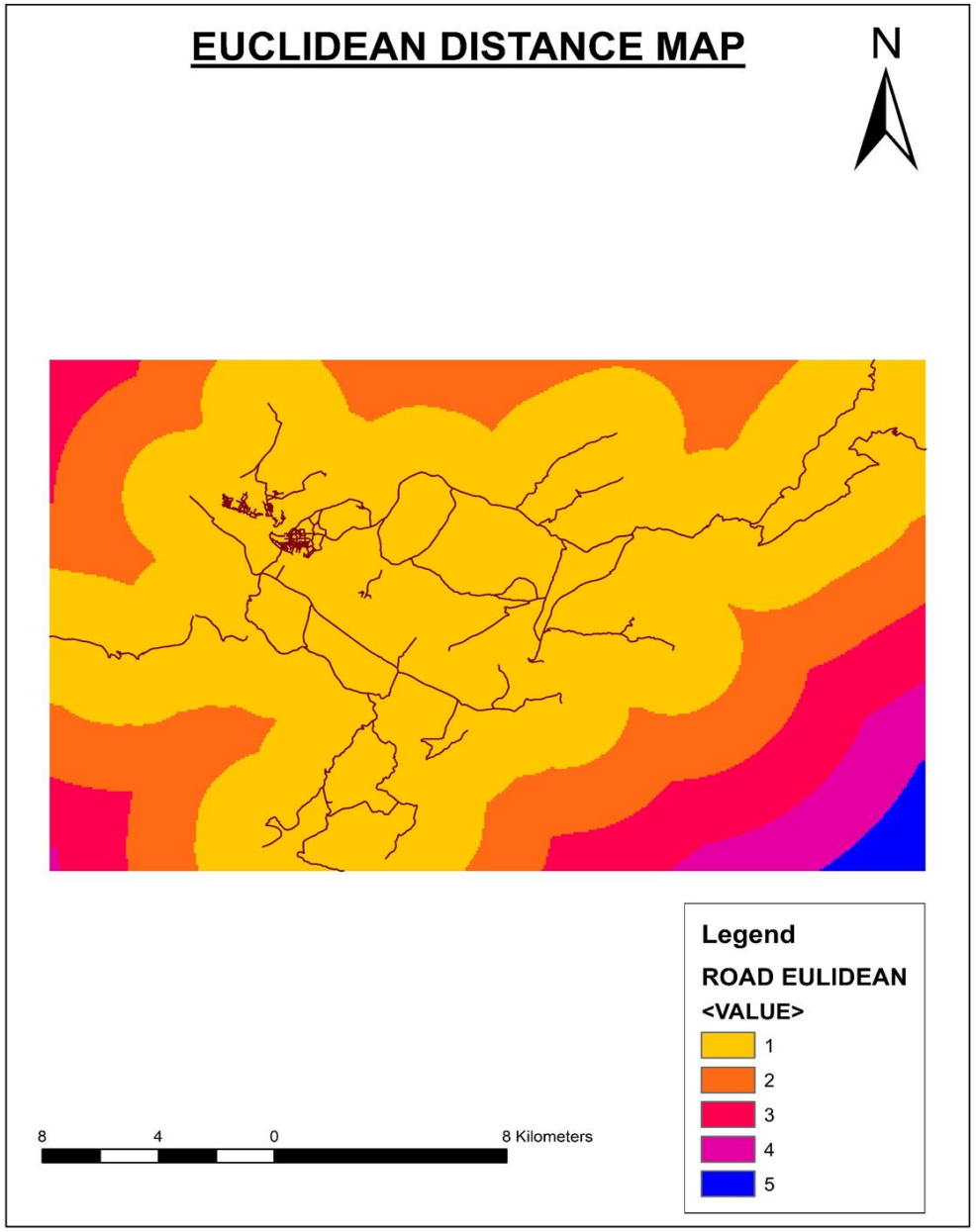
The study of physical features of the surface of the earth and their relation to its geological structure. Geomorphology of a place indicates the landforms present in that particular place.

Data Source: Bhukosh



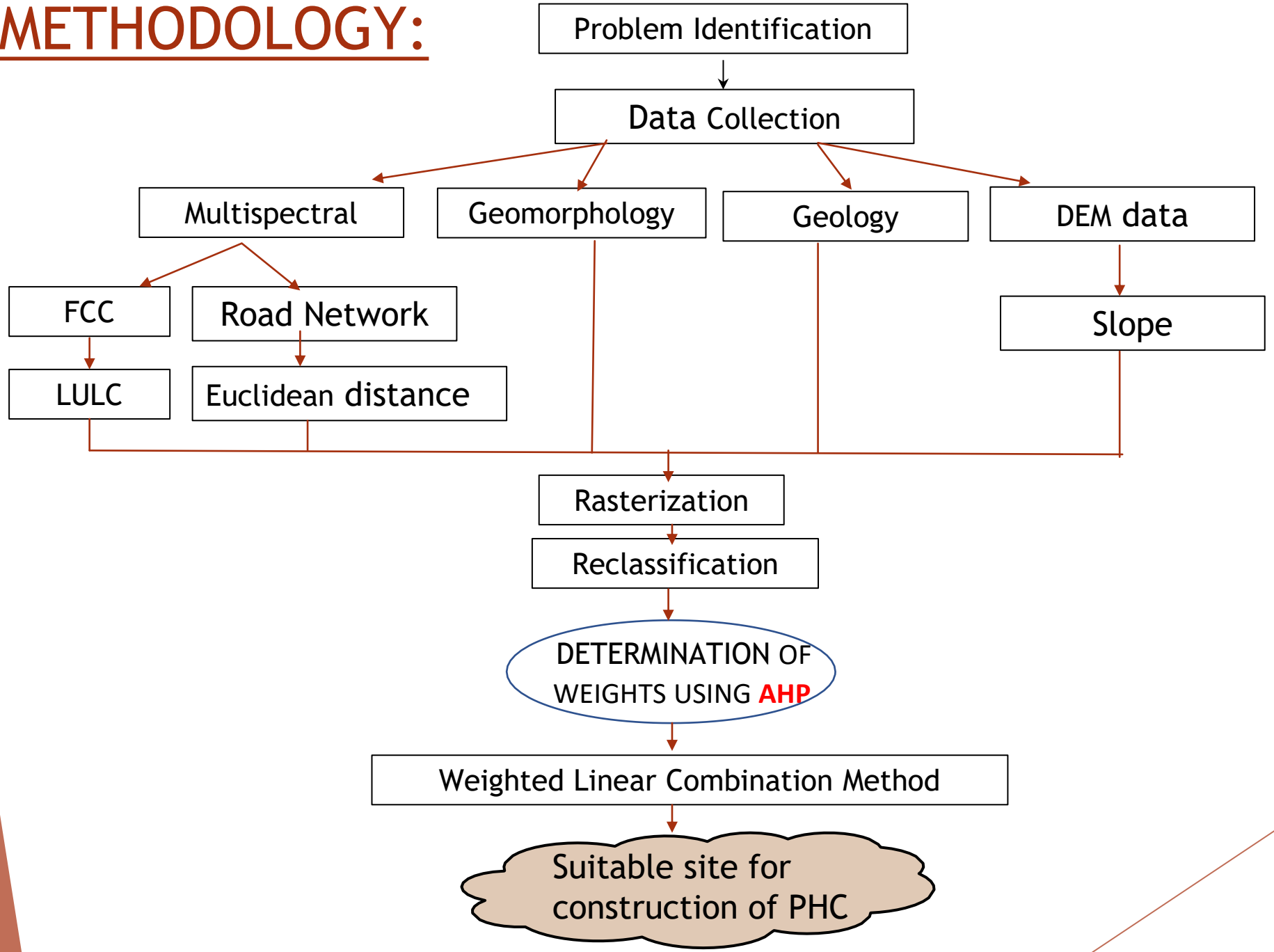
Deals with the study of the structure, evolution and dynamics of the Earth and its natural mineral. The geological structure of araku can be categorized as:
Charnockite> khondalite> migmatite

Data Source: Bhukosh



It measures the distances from every cell to the nearest source. So, by preparing Euclidean distance from major roads display how much a particular site is accessible from the road.
The values mentioned in the map indicates the distances in kms.

METHODOLOGY:



AHP(Analytical Hierarchical Process) :

Is one of the most common used MCE tools. AHP is a method that allows the consideration of both objective and subjective factors in ranking alternatives. Since its introduction in the id 1970s by Thomas Saaty, AHP has been applied in a wide variety of practical applications in various fields including site selection. It assists the decision-making process by allowing decision-makers to organize the criteria and alternative solutions of a decision problem in a hierarchical decision model. The AHP decision hierarchy involves a number of steps: Identification of the goal (e.g., select the most suitable industrial site), use of a set of decision factors/ variables/ criteria (e.g., labour, climate, economic costs, and living conditions), and determination of a set of alternatives/choices (e.g., site 1, site 2 and site 3). The levels of the hierarchy may be expanded as needed (e.g., cost could be considered in terms of labour, utilities, and taxes). At the lowest level on the hierarchy, we find the alternative solutions. Comparisons of the available choices/ alternatives are made on a pair-wise basis.

For example, in considering taxes, AHP would determine whether site 1 is “better” (i.e., has higher tax 3 discount) than site 2 and if so, by how much? Similar comparisons are performed at each level on the hierarchy. This measure of importance/weight is done using a nine-point scale, which is widely utilized in the AHP technique. The AHP process synthesizes the alternatives’ priorities into overall set of values that indicate the relative importance of each factor at the lowest level of the hierarchy.

DETERMINATION OF WEIGHTS FOR CONISIDERED CRITERIA BY AHP:

Steps involved in weight determination are

1. Creating pair wise comparison matrix
2. Normalizing pair wise matrix and weight determination
3. Consistency ratio calculation

Consistency Index (CI):

$$CI = \frac{\lambda_{max} - n}{n - 1}$$

λ_{max}: Principal Eigenvalue

n: dimension of the matrix

$$CR = CI/RI$$

where RI is the average consistency index for random matrices and CI is the consistency index. After all, the priorities of the criteria and sub-criteria are determined based on their weights.

For n=4,RI =0.882

Weighted Linear Combination Method:

Raster Calculation:

w1*reclassified_LULC+w2*reclassified_geomorphology+w3*reclassified_road+

w4*reclassified slope

Preferences for assigning weights:

1. There are six classes in LULC water,forest,barren land,vegetation,sediments and built-up this LULC given more preference to barren land and least preference to water.
2. In geomorphology,three classes namely hill, valley and pediment plane. High preference is given to pediment plane more than valley and least preference is hill.
3. Regarding road network and slope given more preference to slope.

The Fundamental Scale for Pairwise Comparisons		
Intensity of Importance	**Definition**	**Explanation**
1	Equal importance	Two elements contribute equally to the objective
3	Moderate importance	Experience and judgment slightly favor one element over another
5	Strong importance	Experience and judgment strongly favor one element over another
7	Very strong importance	One element is favored very strongly over another; its dominance is demonstrated in practice
9	Extreme importance	The evidence favoring one element over another is of the highest possible order of affirmation
Intensities of 2, 4, 6, and 8 can be used to express intermediate values. Intensities 1.1, 1.2, 1.3, etc. can be used for elements that are very close in importance.		

Computations:

Pairwise Comparison Matrix

	LULC	GEOMOR.	ROAD	SLOPE
LULC	1	4	3	5
GEOMOR.	0.25	1	2	4
ROAD	0.33	0.50	1	2
SLOPE	0.20	0.25	0.50	1
	1.78	5.75	6.50	12.00

Calculation of consistency ratio

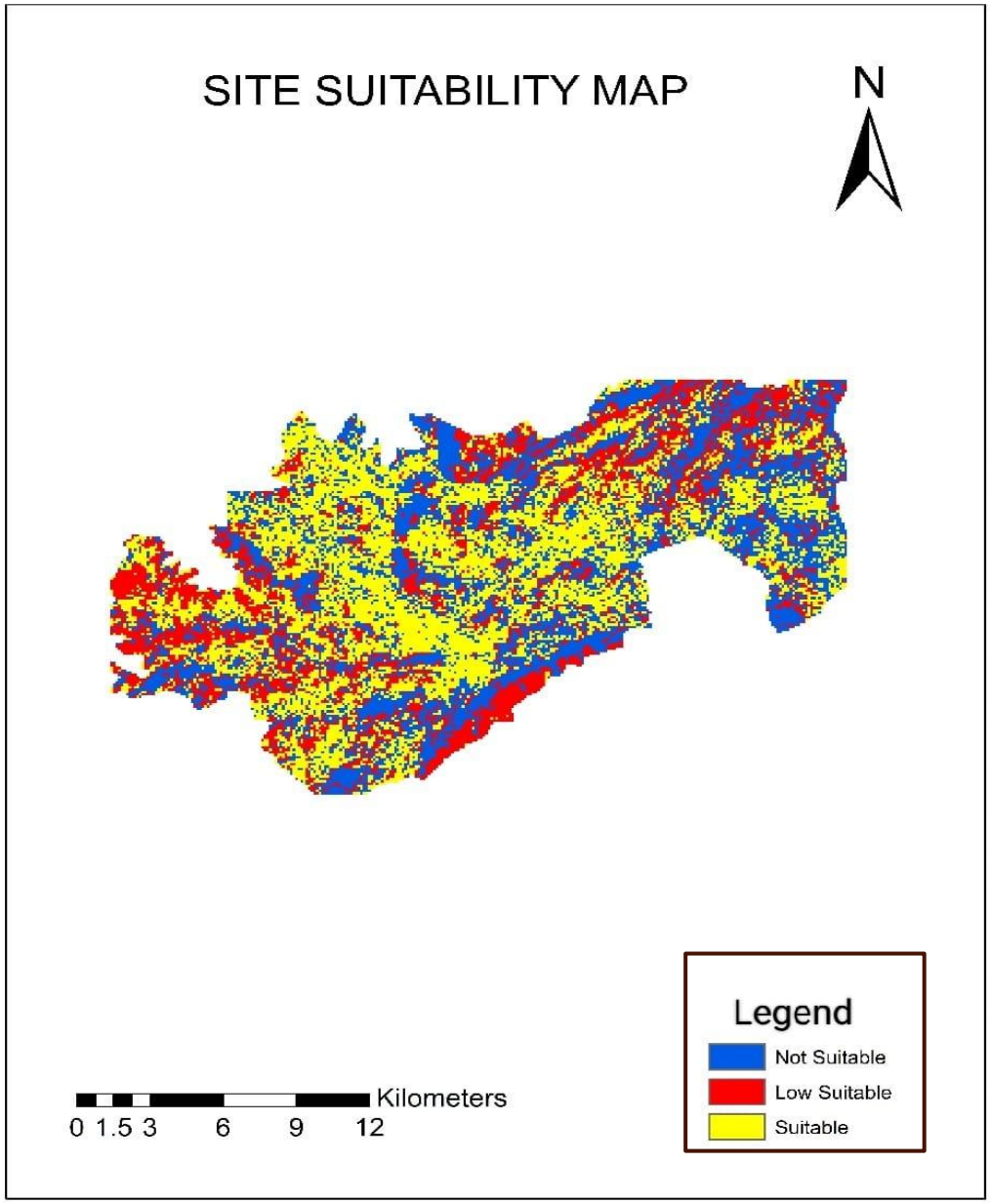
	LULC	GEOMOR.	ROAD	SLOPE	ROW SUM	WEIG.	EIGHEN.V
LULC	0.56	0.70	0.46	0.42	2.13	0.53	0.95
GEOMOR.	0.14	0.17	0.31	0.33	0.96	0.24	1.37
ROAD	0.19	0.09	0.15	0.17	0.59	0.15	0.97
SLOPE	0.11	0.04	0.08	0.08	0.32	0.08	0.95

MAX EV	4.24
CI	0.08
CR	0.09

CR=0.09

Consistency ratio < 0.1 hence, the suitability is consistent

RESULT:


SITE SUITABILITY MAP
N
Kilometers
0 1.5 3 6 9 12
Legend
Not Suitable
Low Suitable
Suitable

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THANK YOU